

3.4 Hooke's law

Further Mechanics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A Hookean spring obeys $T=kx$ for extension x within elastic limit. Elastic energy stored is $\frac{1}{2}kx^2$, the area under its force–extension graph. For systems with two springs, identify whether common force (series) or common extension (parallel) applies before replacing them by an effective stiffness.

Worked example: follow the reasoning

For parallel springs $k_1=20$ and $k_2=30 \text{ N m}^{-1}$, total force at extension x is $(20+30)x$, so $k_{\text{eff}}=50 \text{ N m}^{-1}$.

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A spring of stiffness 200 N m^{-1} stretches 0.10 m . Calculate tension and energy stored. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$T=kx=20 \text{ N}$ [1]; energy= $\frac{1}{2}kx^2$ [1] = 1.0 J [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.